

GRADE 11 CHEMISTRY UNIT 1

ATOMIC STRUCTURE AND PERIODIC PROPERTIES OF THE ELEMENTS

1.1 Introduction

At the end of this section, you will be able to:

- *compare the views of different philosophers on the nature of matter.*

1.2 Dalton's Atomic Theory and the Modern Atomic Theory 3

At the end of this section, you will be able to:

- *state postulates of Dalton's atomic theory*
- *state postulates of the modern atomic theory*
- *state the laws of conservation of mass, definite proportions, multiple proportions and the basis of each of these laws*
- *use postulates of Dalton's atomic theory to explain the laws of definite and multiple proportions*
- *evaluate postulates of Dalton's and the modern atomic theories.*

1.3 Early Experiments to Characterize the Atom 6

At the end of this section, you will be able to:

- *discuss the discovery of the electron*
- *describe the properties of cathode rays*
- *define the terms: radioactivity, radioactive decay and radio-isotope*
- *describe the common types of radioactive emissions*
- *illustrate the alpha scattering experiment, and summarize and interpret the major contribution of experiments of Thomson, Millikan and*
- *Rutherford concerning atomic structure.*

1.3 Make-up of the Nucleus 12

At the end of this section, you will be able to:

- *describe the make-up of the nucleus*
- *define atomic mass*
- *define isotope, and calculate the relative atomic mass (atomic mass) of naturally occurring isotopic elements.*

1.5 Electromagnetic Radiation and Atomic Spectra

At the end of this section, you will be able to:

- *characterize electromagnetic radiation (EMR) in terms of wavelength, frequency and speed*
- *calculate the wavelength and frequency of EMR*
- *explain the dual nature of light*
- *describe emission spectra of atoms as consisting a series of lines*
- *define a photon as a unit of light energy*
- *distinguish how the photon theory explains the photoelectric effect*
- *identify the relationship between a photon absorbed and an electron released*
- *state Bohr's assumption of energy of the electron in an hydrogen atom*
- *calculate the radius of electron orbit, the electron velocity and the energy of an electron using Bohr's model*
- *explain that the line spectrum of hydrogen atom demonstrates the quantized nature of the energy of its electron*
- *explain that atoms emit or absorb energy when they undergo transition from one state to another*
- *compose the limitations of Bohr's theory.*

1.6 The Quantum Mechanical Model of the Atom

At the end of this section, you will be able to:

- *state Heisenberg's uncertainty principle*
- *describe the significance of electron probability distribution.*
- *explain the quantum numbers n , l , m_l , m_s*
- *write all possible sets of quantum numbers of electrons in an atom.*
- *describe the shapes of orbitals designated by s , p and d .*

1.7 Electronic Configurations and Orbital Diagrams

At the end of this section, you will be able to:

- *explain the Aufbau Principle*
- *explain Pauli's Exclusion Principle*
- *explain Hund's Rule*
- *write ground state electron configurations of multi-electron atoms.*

1.8 Electronic Configurations and the Periodic Table of the Elements

At the end of this section, you will be able to:

- *correlate the electron configurations of elements with their positions in the periodic table*
- *give a reasonable explanation for the shape of the periodic table*
- *classify elements as representative, transition and inner-transition elements*
- *explain the general trends in atomic radius, ionization energy, electron affinity, electronegativity, and metallic character of elements within a period and group of the periodic table*
- *write the advantages of the periodic classification of elements.*